

# Tables



1

a

|        |     |   |     |      |
|--------|-----|---|-----|------|
| $x$    | 5   | 6 | 6.9 | 6.99 |
| $f(x)$ | 0.5 | 1 | 10  | 100  |

b

|        |      |    |     |      |
|--------|------|----|-----|------|
| $x$    | 9    | 8  | 7.1 | 7.01 |
| $f(x)$ | -0.5 | -1 | -10 | -100 |

c The limit does not exist.

2

a

|        |       |         |            |            |         |       |
|--------|-------|---------|------------|------------|---------|-------|
| $x$    | 1.9   | 1.99    | 1.999      | 2.001      | 2.01    | 2.1   |
| $f(x)$ | 19.05 | 20.8005 | 20.980 005 | 21.020 005 | 21.2005 | 23.05 |

b  $\lim_{x \rightarrow 2} (5x^2 + 1) = 21$

3

a

|        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|
| $x$    | -0.1   | -0.01  | -0.001 | 0.001  | 0.01   | 0.1    |
| $f(x)$ | 1.0448 | 1.0049 | 1.0005 | 0.9995 | 0.9950 | 0.9453 |

b  $\lim_{x \rightarrow 0} \left( \frac{2-x}{x^2+2} \right) = 1$

4

a

|        |     |      |       |       |      |     |
|--------|-----|------|-------|-------|------|-----|
| $x$    | 3.9 | 3.99 | 3.999 | 4.001 | 4.01 | 4.1 |
| $f(x)$ | 3.9 | 3.99 | 3.999 | 4.001 | 4.01 | 4.1 |

b  $\lim_{x \rightarrow 4} \left( \frac{x^2 - 4x}{x - 4} \right) = 4$

5

a

|        |      |        |           |           |        |      |
|--------|------|--------|-----------|-----------|--------|------|
| $x$    | -1.1 | -1.01  | -1.001    | -0.999    | -0.99  | -0.9 |
| $f(x)$ | 4.31 | 4.0301 | 4.003 001 | 3.997 001 | 3.9701 | 3.71 |

b  $\lim_{x \rightarrow -1} f(x) = 4$

6



a

|        |        |         |          |         |        |       |
|--------|--------|---------|----------|---------|--------|-------|
| $x$    | 4.9    | 4.99    | 4.999    | 5.001   | 5.01   | 5.1   |
| $f(x)$ | -62.14 | -623.38 | -6235.84 | 6236.29 | 623.83 | 62.58 |

b No, because the function is not approaching the same value on either side of  $x = 5$ .

7

a

|                         |        |          |        |    |        |         |        |
|-------------------------|--------|----------|--------|----|--------|---------|--------|
| $x$                     | -5.1   | -5.01    | -5.001 | -5 | -4.999 | -4.99   | -4.9   |
| $\frac{x^2 + 4}{x + 5}$ | -300.1 | -2910.01 | -29010 | -  | 28990  | 2890.01 | 280.01 |

b No, because the function is not approaching the same value on either side of  $x = -5$ .

8

a

|                                                |          |          |        |   |        |          |          |
|------------------------------------------------|----------|----------|--------|---|--------|----------|----------|
| $x$                                            | 2.9      | 2.99     | 2.999  | 3 | 3.001  | 3.01     | 3.1      |
| $\left( \frac{e^{x-3} + x - 4}{x - 3} \right)$ | 1.951 63 | 1.995 02 | 1.9995 | - | 2.0005 | 2.005 02 | 2.051 71 |

b Yes, because the function is approaching the same value of 2 on either side of  $x = 3$ .

## Graphs

9

a  $y\text{-value} = 2$

b  $y\text{-value} = -1$

c No, because  $g(x)$  approaches 2 on the left of  $x = -2$ , and  $g(x)$  approaches  $-1$  on the right of  $x = -2$ .

10

a  $y\text{-value} = -2$

b  $y\text{-value} = -2$

c The limit of  $\frac{x^2 - 1}{x + 1}$  as  $x$  approaches  $-1$  is  $-2$ .

d  $\lim_{x \rightarrow -1} \left( \frac{x^2 - 1}{x + 1} \right) = -2$

11 No, because the function approaches different values on either side of  $x = 3$ .



12 No, because the function approaches different values on either side of  $x = 0$ .

13 Yes, because of either side of  $x = 5$ , the function approaches infinity as  $x \rightarrow 5$ .

14 Yes, because on either side of  $x = 0$  the function approaches 2.

15  $\lim_{x \rightarrow 3} f(x) = 1$

16  $\lim_{x \rightarrow 0} f(x) = 0$

17  $\lim_{x \rightarrow 3} f(x) = 5$

18  $\lim_{x \rightarrow 3} f(x) = 4$

19  $\lim_{x \rightarrow 2} f(x) = 1$

20

a  $\lim_{x \rightarrow -4^+} \left( \frac{1}{x+4} \right) = \infty$

b  $\lim_{x \rightarrow -4^-} \left( \frac{1}{x+4} \right) = -\infty$